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Mark Dittner , and Bernd Hartke 



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Mark Dittner^{a)} and Bernd Hartke^{b)}

AFFILIATIONS

Institute for Physical Chemistry, Christian-Albrechts-University Kiel, 24098 Kiel, Germany

^{a)}Author to whom correspondence should be addressed: dittner@pctc.uni-kiel.de

^{b)}Electronic mail: hartke@pctc.uni-kiel.de

ABSTRACT

In a previous paper [M. Dittner and B. Hartke, J. Chem. Theory Comput. **14**, 3547 (2018)], we introduced a preliminary version of our GOCAT (globally optimal catalyst) concept in which electrostatic catalysts are designed for arbitrary reactions by global optimization of distributed point charges that surround the reaction. In this first version, a pre-defined reaction path was kept fixed. This unrealistic assumption allowed for only small catalytic effects. In the present work, we extend our GOCAT framework by a sophisticated and robust on-the-fly reaction path optimization, plus further concomitant algorithm adaptations. This allows smaller and larger excursions from a pre-defined reaction path under the influence of the GOCAT point-charge surrounding, all the way to drastic mechanistic changes. In contrast to the restricted first GOCAT version, this new version is able to address real-life catalysis. We demonstrate this by applying it to the electrostatic catalysis of a prototypical Diels–Alder reaction. Without using any prior information, this procedure re-discovers theoretically and experimentally established features of electrostatic catalysis of this very reaction, including a field-dependent transition from the synchronous, concerted textbook mechanism to a zwitterionic two-step mechanism, and diastereomeric discrimination by suitable electric field components.

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I. INTRODUCTION

From the viewpoint of the potential energy surface (PES) paradigm, discovering new chemical substances and reactions is equivalent to the task of finding new minima on the PES and/or new connections [transition states (TSs) and reaction paths] between them. In recent years, theoretical and computational chemistry has made great strides in these directions [see Refs. 1–4 and references cited therein]. Given the vast size of chemical space and the tight resource limits restricting laboratory-based PES exploration power, these new computational tools will rapidly become very important to discover new molecules, materials, and reactions.

A different route toward new chemistry is to modify the PES itself, specifically toward lower energy barriers along reaction paths, commonly known as catalysis. Despite great long-time efforts being invested in homogeneous, heterogeneous, and bio-enzymatic catalysis,^{5–10} research in these areas is still driven more by serendipity than by rational and/or computational design. However, the huge importance of catalysis is extremely clear not only for getting access to desired reaction products at less cost (as in the shown example

of nitrogen fixation via the Haber–Bosch process at elevated temperatures and pressures vs via nitrogenase enzymes under ambient conditions, cf. Ref. 11 and references therein) but also for enabling control over intricate multi-reaction networks, as again ubiquitously realized in biochemical systems.¹²

Given the importance but also the great challenge of computational catalyst design, we think that it will prove useful to break the problem into smaller subproblems. First of all, we step back from full, multistep catalytic cycles to a single step within such a cycle (for example, the rate-limiting one). For this step, we refrain from including the initial association between the catalyst and substrate (if it is part of this step) and, similarly, the final dissociation between the catalyst and processed substrate; instead, the catalyst and substrate are assumed to be spatially associated from the beginning to the end. This narrows the remaining task down to lowering of the energy profile for this elementary catalytic reaction. Nevertheless, this is still a hard task, given the vastness of chemical space and the total lack of *a priori* information on which catalytic molecules could bring about the desired energy-lowering. Hence, we finally split this remaining task into two simpler steps, namely, (A)

designing an optimal but abstract catalyst followed by (B) finding concrete realizations (molecules, surfaces, and cages) for it. This two-step procedure has several substantial advantages compared to performing both steps at the same time: In step (A), the vast search space of chemical possibilities can be replaced by relatively small search spaces of well-defined, abstract interaction entities (electric fields, steric restrictions, donors/acceptors of hydrogen bonds, protons, electrons, hydrogen atoms, etc.). In contrast to concrete catalyst exploration (as in Ref. 13 where machine learning and computational alchemy were employed to screen metal atom transmutations in prescribed geometries of organic molecules on metal slabs), this deliberate abstraction in step (A) makes it practically possible to employ global optimization across a search space of all possibilities, finding a theoretically maximal catalytic effect. This is then a guideline for step (B), where concrete catalyst design necessarily has to be performed within an *a priori* restricted subsection of chemical space. Knowledge of the abstract maximal catalytic effect allows us to decide if a globally optimized, concrete catalyst obtained from examining one subsection of chemical space is good enough or if a new search should be made in another (possibly very different) subsection. Additionally, step (A) automatically produces mechanistic insight into how (optimal) catalysis works for the reaction under study. This can be very useful even if step (B) is then performed not by computational means but by design arising from traditional human deliberation. As a side note, it is also possible to turn this question around and optimize better substrates for a given catalyst, which is a highly relevant research area in protein biochemistry. Also in this case, the problem has been split in two steps, with the first one being an optimization of partial charges, as in Ref. 14.

Parallel to similar work by other groups^{15–21} (for a recent review, see Ref. 21 and references therein), in previous work,²² we have already established a restricted approach to step (A). We introduced the general framework of globally optimal catalysts (GOCATs). Around the collective, superimposed reaction-path snapshots of a given reaction, additional interaction entities of differing representations and abstraction levels are placed and then globally optimized to achieve maximal lowering of the reaction barrier.

One of the simplest but arguably the most important of such embeddings are partial point charges (Coulomb centers), generating an additional electrostatic potential (ESP) and electric field (EF) that influence the target reaction. Hence, this GOCAT type falls into the regime of general *electrostatic catalysis*. Apart from the well-known pioneering work,²³ this electrostatic catalysis has become very popular in recent years and has been recently reviewed in both the theoretical^{24–26} and the experimental domains.^{27,28} In this area, the so-called oriented external electric fields (OEEFs)²⁴ are employed, which are usually dipolar uniform fields along a fixed given direction [or the ones produced at scanning tunneling microscope (STM) tips or charged surfaces].²⁵ In the cited reviews, it was demonstrated for different reactions (e.g., Refs. 29–35) that an OEEF along the direction of electron reorganization catalyzes even *nonpolar* (yet polarizable) and *non-redox* reactions by several orders of magnitude. Shaik *et al.* named this direction the *reaction axis*, the direction in which the “electrons move for building bonds” during the formation of the products from the reactants. Fields in other directions can then control regio- and stereoselectivity. With the

increase in the field strength, this impact can induce a mechanistic crossover with emerging ionic species. Most of these influences can be understood by a field-induced stabilization of ionic structures. Indeed, this recent research regarding electrostatic catalysis due to OEEFs puts forward *abstract* embeddings for pure electrostatic effects,^{24–26} which is in line with our current GOCAT model. The importance of the orientation of an external electric field along a reaction axis is also supported by experiments. Reference 36 proved that a correct orientation of the external electric field (along or at least non-orthogonal to the reaction axis) is indeed crucial for proper electrostatic catalysis.

Similarly, in the theoretical domain, local projections of EFs onto the important bonds were often applied for quantifying the effects of electrostatic surroundings onto given reactions.^{27,37–41}

This view also extends to enzymatic catalysis,^{31,38–43} which leads back to the proposal of Warshel that enzymes work via electrostatics (cf. Refs. 23 and 44–46). However, a vivid debate of this issue should be emphasized: The proponents of *electrostatic* enzymatic catalysis focus on a preorganized polar enzymatic environment stabilizing the transition state more than the reactants, while others argue for an (additional) importance of *dynamical effects*. Warshel most vigorously presented the first (electrostatic) view, as argued in detail in different studies (Refs. 46–48) against the other view (see Refs. 49–51, the introduction of Refs. 27 and 41, and references therein).

At any rate, with the GOCAT model, we do not want to take sides in this debate and we do not need to do so: If an electrostatic preorganization sufficed to describe the full power of enzymes, this would doubtlessly highlight the importance of introducing and optimizing (abstract) point charge GOCATs. If not, this would stress the fact that naturally occurring catalysts need more specific dynamical effects and so should a further extended GOCAT model. Note, however, that in its current stage, our GOCAT model still is less flexible than it could be: We optimize one single distribution of non-polarizable partial charges for the whole reaction path, which implicitly excludes changes in this distribution induced by the substrate. These may be classified as missing “dynamic” effects.

Electrostatic fields within enzymes can now be measured by vibrational Stark shift spectroscopy. This was put forward and discussed in Refs. 27, 52, and 53 by Fried and Boxer. With the help of additional computational calibration within solvatochromism studies, even absolute scales of electric fields can be deduced in this way. For instance, EFs up to $\sim 150 \text{ MV cm}^{-1}$ (or $34.6 \text{ kcal mol}^{-1} \text{ \AA}^{-1} \text{ e}^{-1}$ in the units used later on) were measured in the enzyme ketosteroid isomerase in the active site.⁵² However, such EFs in organized matter (cocrystals) are seemingly not that “extreme” but rather the common case ($\sim 100\text{--}190 \text{ MV cm}^{-1}$), which was experimentally measured in high-resolution diffraction experiments and inferred from the charge density by Shi *et al.* in Ref. 54. Moreover, Ciampì *et al.* have investigated the role of EFs and how to harness them for catalysis *in practice*, reaching even the daily laboratory regime.²⁸ In a single-molecule setting, electrostatic catalysis for a Diels–Alder (DA) reaction was experimentally proved to occur using an STM “blinking” experiment.⁵⁵ This has shown *qualitative* agreement with the computational studies in the same reference. As STM is undeniably not a “high-throughput” approach for EF catalysis, “through-space” EF effects of functional groups at the molecular systems have also been investigated further. Klinska and co-workers showed, for

instance, H-atom transfer reactivity enhancements due to local electric fields from pH-switches⁵⁶ and a high catalytic influence on a DA reaction, also including *endo/exo* selectivity,⁵⁷ under practical solution-phase conditions.

With our purely electrostatic GOCAT model, as introduced in Ref. 22, we were able to study (arbitrary) ESP/EF impacts on chemical reactions. By (only) setting boundary conditions such as the number of total partial charges, N_{Ch} , ranges of the charge magnitudes, $q_i \in [q_{\text{min}}, q_{\text{max}}]$, and a common embedding surface (e.g., on a van der Waals surface, vdW), *globally* optimal (i.e., *maximal*) catalytic effects can be determined by *global* optimization without any further explicit bias besides the mentioned boundaries. We would like to emphasize that our electrostatic GOCAT approach encompasses both the uniform EFs usually considered in electrostatic catalysis (cf. references cited above) and also highly nonuniform (nonhomogeneous) ones that may be closer to EFs in enzymatic catalysis. Nevertheless, the final interpretations of GOCATs for a neutral S_N2 reaction in Ref. 22 were perfectly in line with the aforementioned “reaction axis” rule and also with the independent results of Ref. 35 for the same reaction.

Still, a couple of crucial approximations were present in the GOCAT model of our previous work²² where we discussed them in detail. While yet sticking to a purely electrostatic surrounding, the most critical assumption is clearly that of a fixed/pre-set reaction path for the reaction to be energetically manipulated. From the outset, this excludes any subtle shifts, further relaxations, or extensive mechanistic changes that may be conceivably induced by the GOCAT itself.

Therefore, in the work presented here, we generalize our earlier approach²² by now relaxing this key restriction. In essence, we combine catalytic PES manipulation by our evolving GOCATs with on-the-fly reaction-path re-optimization. We show here that in this generalized setting, our GOCAT concept still works in the sense of delivering catalytically optimally enhanced reaction profiles but that now, also new reaction paths can be found arising from the GOCATs during their optimization and substantially differing from the ones put in originally. Clearly, this quasi-self-consistent feedback loop is more reasonable from a method design viewpoint, but it also comes closer to what a chemist expects from catalysis since (as we show below) it now also allows qualitative changes in the reaction profile toward the appearance of intermediates and more than one energy barrier between them.

In contrast to our earlier work²² where we applied GOCATs to a simple S_N2 -style Menshutkin reaction, here, we chose a DA reaction as the target for several reasons: First, it is a more complicated reaction type enabling more interesting catalysis. Second, it allows us to directly relate to the broad literature regarding DA, both of a more general nature (see Ref. 58 and similar works), and the above-cited EF-catalysis studies of DA reactions;^{32,34,55,57} this provides us with the possibility to test and rationalize our GOCAT predictions. Last but not least, DA reactions are a natural choice⁵⁹ at Kiel University.

The remainder of this article is organized as follows: In Sec. II, the global optimization methodology is briefly sketched, with emphasis on the modified objective function that incorporates the reaction path relaxation, and the GOCAT boundary conditions of the models are defined. Selected main results based on this new objective function are then shown and discussed in Sec. III, which, first of all, also introduces the Diels–Alder reaction of the

application. We end with a conclusion in Sec. IV. Note that many more results and details are given in the [supplementary material](#) of this publication.

II. IMPLEMENTATION AND METHODS

For completeness, we briefly discuss the most important technical details in the remainder of this section. To skip these technical details, read the brief summary in the next paragraph and then skip forward to Sec. III.

As discussed in Sec. I, the key aim is to find a distribution of partial charges around a given reaction as an abstract catalyst representation. This partial charge distribution is globally optimized using our genetic algorithm package OGOLEM.⁶⁰ In contrast to our earlier work,²² we allow arbitrary reaction-path changes in the presence of this partial charge distribution by performing on-the-fly reaction-path re-optimizations with the nudged-elastic-band algorithm.^{61,62} As a fitness indicator of the resulting GOCAT models, we essentially take the barrier height reduction; additional refinements [including reactant and product (de)stabilization] are discussed in the following subsections. The partial charges are either present on the joined van der Waals surface of the substrate reaction path frames (as before²²) or on a sphere enclosing it, with the latter leading to less intuitive charge placements but also to simpler and hence easier understandable EFs. To relate the existing literature regarding OEEFs for this DA reaction to the present results, we also employ purely homogeneous EFs (built from partial charges arranged on “plate capacitor” planes on two opposite ends of the substrate frames without any optimization of these charges) and use the “reaction axis” concept in the analysis of our results.

A. Global optimization

We use genetic algorithms (GAs)^{63–65} as metaheuristic⁶⁶ for the global optimization of GOCATs, as implemented for various chemical problems in our OGOLEM framework.⁶⁰ OGOLEM employs a generation-free⁶⁷ GA for advanced scaling, failsafely parallelized across homo-/heterogeneous hardware^{68,69} (i.e., “immune” to hardware failure) with a plethora of different mutation and crossover operators,^{22,70–72} ranging from abstract ones to very problem-specific ones, and includes niching^{73,74} as the main diversity conservation protocol. The modifications to OGOLEM for optimizing the positions and charge values of the surrounding GOCAT on specific surfaces around/close to chemical reactions were already discussed in Ref. 22. One further wrapper for crossover/mutation operators was added, which introduces ESP-aware steps/leaps through the search space. After any other operator, such as a bigger random mutation step in Cartesian space, the resulting GOCAT is directly re-optimized by this wrapping operator in order to result in a similar ESP at the core reacting atoms that were present already before the operator application. In this way, even more extended steps in the search space can take place for specific sampling of variations of similarly composed ESP/EF embeddings. The reason behind this is that after some first GA progression, when already chemically meaningful GOCATs have evolved, this ESP information is re-used for an educated guess to sample alike GOCATs in the ESP domain but different with regard to its intrinsic coordinates (Cartesians and charge values) and to increase the efficiency for subsequent

GA operations in the same overall global optimization. As always, we use a weighted mixture of each of these mutation/crossover and wrapper operators as a protocol for minimal bias throughout the GA optimization. However, these settings were greedily benchmarked for a rough sketch of increased performance for the problem at hand.

B. Reaction path optimization

For the reaction path search in the form of the minimum energy path, we use an implementation of the nudged elastic band (NEB)^{61,62} methodology in *OGOLEM*, mostly optimized using the fast inertial relaxation engine (FIRE).⁷⁵ This was already implemented and discussed in our paper regarding the framework.²² Moreover, to make the whole NEB relaxation more robust as well as to save some energy/gradient evaluations, i.e., to make the optimization more computationally efficient, we follow the delineation of the *automated* NEB version developed in Ref. 76. In this case, the chain of states/frames between the end frames, i.e., the reactants and products, are usually not created and optimized in *one* batch. Instead, a rough NEB is first created, which consists of just a few frames, and then, additional ones are added successively at the places where they are needed. The decision about the spot to add another frame, i.e., to increase the resolution of the minimum energy path (MEP), is based on geometrical and energetic resolutions. Final criteria for a maximum allowed threshold of geometrical gaps between the frames in the final MEP can also be incorporated in this way. Thus, the total number of frames reached at the end of the complete NEBs for one MEP optimization will be determined by this resolution and not by a pre-fixed number. The subsequent NEB optimization with a new frame is usually window-based, i.e., it addresses the newly added frame plus some of its neighbors but not all the frames of the antecedent NEB runs. This will be called “sub-NEB” below. The automated version with adaptive frame additions is especially beneficial when the MEP on the PES happens to be much more curvilinear or different from the interpolated starting path between the end frames. For further details, see Ref. 76 as we strictly follow the depictions mentioned there for our implementation. All the interpolations done—at the beginning of the first NEB or between each subsequent window-based sub-NEB—are generated as another (completely independent) NEB on the so-called image dependent pair potential (IDPP), as introduced in Ref. 77 (for a further discussion, also compare with Ref. 78). Initial MEPs in the gas-phase are made translationally and rotationally invariant via the center of mass transformations and optimal rotations based on quaternions using Kearsley’s algorithm,^{79,80} also discussed in Ref. 81. For NEBs within GOCATs, when the surrounding ESP/EF is generated due to the partial charges, the GOCAT serves as an anchor point and no such transformations are needed anymore.

C. Objective function

We will abbreviate reactant(s) as R, the transition state as TS, and the product(s) as P consistently in the remainder of this publication. The quality assessment of each GOCAT is performed by the objective function—which will be treated as being synonymous—or the fitness function. This function includes multiple different objectives, all of which are accumulated into one strictly positive number

to be globally, metaheuristically optimized during the GA within *OGOLEM*. Note that this function was explained in detail in the first paper of the present authors regarding the framework, especially in the [supplementary material](#) of that publication (including typical and yet arbitrary weights).²² In short, each gradient at the pristine stationary points (R, TS, and P) is checked whether it lies inside a (usually rough) gradient window. Additionally, the unimodality of the connected reaction energy path is enforced, meaning that the emerging reaction profiles increase monotonically from R to TS and decrease monotonically from TS to P. Also no discrete shifts of the extremal energetic points are allowed. If these criteria are not fulfilled, additional fitness values (as penalties) are added as a mixture of continuous and discrete terms. On the other hand, if these criteria are fulfilled, the final fitness is essentially the final energetic reaction barrier—which is to be minimized as the target for catalysis. For convenience, this is illustrated again in algorithm S1 (see the [supplementary material](#)). Obviously, *this* setting only allows us to optimize a lower barrier for a pre-defined mechanism between (at least) pre-defined R and P structures, i.e., *vertically* (or statically) in energy on the PES. In other words, only *one* reference list of structures for the reaction path was “frozen” and energetically optimized to show the intended properties.

The main purpose of the present paper is to extend this methodology in order to include mechanistic changes during the GOCAT optimization of the treated reaction and to lift some of the arbitrary criteria, such as the rough gradient norms. To this end, an *adaptive* (or non-vertical) fitness function is illustrated in Algorithm 1.

This can be summarized into the following steps:

- If the incoming GOCAT to be evaluated is within the catalytic region (see line 3 of Algorithm 1), which is set to be the fitness of the pristine reaction without any catalysis, a reoptimization of the MEP is done. Otherwise, the GOCAT is evaluated only with the static fitness function. This hard check is performed in order to filter out many highly unchemical results, which, in most of the cases, would not lead to any sensible catalysis even after the MEP optimization (or only to numerical problems or the failure of the latter).
- Lines 4–6: the MEP optimization starts with a relaxation (local energy minimization) of the two end frames, i.e., R and P. As some numerical problems can still occur due to a GOCAT which does not allow to result in a sensible (stable) structure within the electrostatic embedding, multiple sanity checks are always applied. For the end-point relaxation step, here, the gradients are checked again to be lower than a tight gradient norm threshold. (Note: within exotic or unchemical embeddings that happen to be sampled during the GA in the case of only including additional Coulomb centers, *concave* energetic regions on the PES also could result, having no proper stationary point: That is, an atom is attracted too strongly to a point charge such that the gradient *increases* in a negative gradient direction. Thus, such broken GOCATs must be sorted out this way and, moreover, the local optimization must be robust.)
- Lines 7–12: with proper new end-frames, i.e., minima structures for R and P, the NEB starts. Both for performance and robustness, the automated (sub-)NEB version discussed

ALGORITHM 1: Adaptive fitness function for GOCAT design.**Input:** GOCAT: g **Data:** parameters for adaptive NEB & FIRE; weights/ingredients of for
fitnessFunction($\cdot$) (see Algorithm S1, ESI)**Result:** GOCAT surrounding new MEP and its fitness**1 Function adaptiveFitnessFunction(g):**

```
/* relax GOCAT and calculate fitness */
2  $f \leftarrow \text{fitnessFunction}(g)$ 
3 if  $f \leq \text{threshold}$  then // fitness in catalytic region
4    $g' \leftarrow \text{relaxEndPoint}(g)$  // end-frames are locally optimized
5   if  $\text{sanityCheck}(g')$  not sane then // check if  $\|\nabla E\|$  small enough
6     return  $g$  (unchanged) with  $f$  as fitness
7    $g' \leftarrow \text{subNEB}(g')$  // non-linearly interpolated NEB between end-frames
8   while  $\text{resolutionCheck}(g')$  not fine do // check the geometric/energetic
    resolution, with an upper bound on the inter-frame distance at the end
9      $g' \leftarrow \text{addFrame}(g')$  // add one further frame via geom./energ. resolution
10     $g' \leftarrow \text{subNEB}(g')$  // window-based NEB of the new and some neighbor
    frames
11  if  $\text{sanityCheck}(g')$  not sane then // sane if no kinks &  $\|\nabla E\|$  small enough
12    return  $g$  (unchanged) with  $f$  as fitness
13   $g' \leftarrow \text{CINEB}(g)$  // CI NEB for TS optimization
14  if  $\text{sanityCheck}(g')$  not sane then // see comment in Line 11 above
15    return  $g$  (unchanged) with  $f$  as fitness
16   $g' \leftarrow \text{TSFixedNEB}(g')$  // tight NEB optimization with fixed TS
17  if  $\text{sanityCheck}(g')$  not sane then // see comment in Line 11 above
18    return  $g$  (unchanged) with  $f$  as fitness
    /* now  $g'$  is fine: final fitness evaluation */
19   $f' \leftarrow \text{fitnessFunction}(g')$ 
20  return  $g'$  (surrounding new MEP) with  $f'$  as fitness
21 return  $g$  (unchanged) with  $f$  as fitness
```

above is used. Frames are separately added to the current MEP, and an additional NEB within a window of the current total frames is optimized, until sufficient geometric and energetic resolution is reached. Between each of these steps, again some sanity checks are needed.

- Lines 13–15: after having optimized an estimate of the TS on this MEP, a climbing image (CI) NEB is started (again window-based, i.e., a sub-NEB).
- Lines 16–18: one further NEB with tight thresholds and including each frame is conducted to compute a precise discrete representation of the MEP.
- Lines 19–20: finally, the fitness is evaluated as described in Ref. 22 and in the [supplementary material](#) of the present paper. However, all criteria referring to gradient norms, shift of frames, etc., are all *per constructionem* not present anymore after this new MEP influenced by a GOCAT is found. Only the final reaction barrier determines the final fitness essentially. Note that strict unimodality (which is/was usually enforced) was also not present in this setting in order to include two-step mechanisms in the search space.

D. GOCAT models and computational settings

Multiple GOCAT boundary configurations were examined but only a subset of these will be discussed in the following. We looked at vdW-based, spherical and plate GOCATs, with either the static/vertical objective function (no MEP relaxation) or the adaptive objective function (full MEP relaxation). All models always enforced charge neutrality of the total GOCAT, $\sum_i q_i = 0$. In Sec. III A, the static vdW and spherical models will be discussed, with corresponding figures given in the [supplementary material](#). In Sec. III B, only the spherical model and the adaptive objective function are used (due to more robust NEB optimizations in the spheres). The plates are only employed as internal reference for fully homogeneous fields and are not optimized in any way. They merely serve to discuss the homogeneous field contribution and to compare with the literature (which also do not optimize the fields). These results are also moved to the [supplementary material](#).

vdW-model. $N_{\text{Ch}} = 10$ partial point charges were optimized directly on a vdW surface of the complete DA reaction path (cf. Sec. III for the description of the DA reaction). The path was generated as the embedding of a fully aligned 18-frame representation of the MEP (for a static/vertical fitness function usage). Furthermore, the minimal distance enforced between the charges i and j was set to $r_{ij} \geq r_{\text{min}} = 1.0 \text{ \AA}$. This vdW-GOCAT will deliver a more inhomogeneous EF around the atoms of the reactions by addressing separate atoms by separate point charges.

Spherical model. In contrast, another GOCAT model with $N_{\text{Ch}} = 81$ is bound to a sphere with a radius of $r = 7.5 \text{ \AA}$ around the geometric center of all atoms in all frames, and a larger inter-charge distance of $r_{ij} \geq r_{\text{min}} = 3.0 \text{ \AA}$ is enforced. This model is intended to deliver a more homogeneous field simply because these spherical GOCATs have a significantly higher distance of at least about $4 \text{ \AA}$ to all embedded reaction atoms and thus cannot have such a local impact as in the vdW case in which EF features almost address single reaction atoms. These spherical GOCATs are also advantageous for the reaction path relaxation during the GOCAT optimization by

the adaptive fitness function, since there is more collision-free space in the center of the GOCAT.

Plate GOCAT model. Moreover, to extract the literature-known simple catalytic trends of completely uniform electric fields along any of the three Cartesian axes to be defined below, GOCATs constituting “parallel plate capacitors” were also evaluated. Here, $N_{\text{Ch}} = 2 \cdot 30 \cdot 30 = 1800$ charges defined on a uniform grid of a quadratic area of $20 \cdot 20 \text{ \AA}^2$ and a distance of $20 \text{ \AA}$ between the plates from the center of the reaction were sampled by a packing operator without any further subsequent GA optimizations. These are then analyzed consistently using the same methodology for all the other GA-optimized GOCAT models. Here, both fields in all three Cartesian directions were generated, which led to either inhibition or catalysis of the DA reaction. For each separate plate GOCAT, the charge value, q_i , is the same for each charge within a plate and of opposing sign between the two plates and was successively changed to create increasing field strengths from small ones up to very extreme ones. One arbitrary example is illustrated in Fig. S21 (see the [supplementary material](#)). A similar approach was recently presented in Ref. 82.

Fitness function variants. Compared to our first framework paper,²² the presently used catalysis of a DA reaction will show more subtle effects. The field can affect the R frame and the TS differently, i.e., due to bigger movements and changes in the molecular dipole moment during the reaction from R to TS to P, the frames can be separately addressed by one surrounding GOCAT. Thus, even GOCAT candidate solutions showing severe reactant destabilizations without TS stabilizations exist in the search space, contrary to Ref. 22, where the high symmetry of the S_N2 reaction studied there ensured that everything was happening along one central attack line. Therefore, the fitness function for DA was used in two different modes that directly translate to two qualitatively different GOCAT optimizations: Either the R frame is enforced to be stabilized, $\Delta E_R \leq 0 \text{ kcal mol}^{-1}$ with $\Delta E_{\{\text{R,TS,P}\}} = E_{\{\text{R,TS,P}\}}^{\text{GOCAT}} - E_{\{\text{R,TS,P}\}}^{\text{ref}}$, by using an additional penalty that deteriorates worse candidate solutions during the search, or without such a penalty. This penalty was employed to mainly “zoom” into the specified basin of attraction of the search space of GOCATs, after we had noticed that many R-destabilizing solutions survive. For the DA reaction at hand, we wanted to focus on the more interesting TS-stabilizing solutions, which can be better worked out by applying this penalty (i.e., to force the stabilization) in the static fitness function. Note also that the adaptive fitness function with MEP relaxation usually leads to the stabilization of the specific frame(s) of this DA reaction since these are then allowed to re-align in the external field, which is not the case in the static fitness function. Moreover, in other reactions (e.g., enzymatic ones), one could also denote R-destabilization as barrier-decreasing effect, and also, one could include explicitly the P side for proper product release. These would constitute further suitable options of objective function variants in future applications.

GA setting. The *exo* and *endo* diastereomers of the DA reaction were treated separately. For each GOCAT setting, up to 6000 individuals were generated, arising from multiple separate GA runs for $2\text{--}3 \cdot 10^6$ iterations that were locally optimized only after the

GA. For the evaluation, we will mainly show some statistical correlations and ESP/EF analyses. Yet, the whole methodology described in Ref. 22 was also used, including the clustering procedures. Some snippets of these will only be given in the [supplementary material](#) of this publication. All calculations shown here employed the PM7⁸³ semi-empirical Hamiltonian as implemented in MOPAC⁸⁴ in order to achieve enough computational efficiency for the global optimization, and the Coulomb centers were treated as “MM” part in the QM/MM approach implemented there.⁸⁵

III. RESULTS AND DISCUSSION

The DA reaction to be catalyzed in the following is presented in Fig. 1. Both the *exo* and *endo* diastereomers of the [4 + 2] cycloaddition reaction of cyclopentadiene and maleic anhydride are illustrated. Since the dienophile is an anhydride here, the reaction is already slightly accelerated compared to the plain “barebone” reaction, the reaction without any such functional groups, i.e., between butadiene and ethylene. This is also indicated by the partial charges, $\delta^{(+,-)}$, as used in organic chemistry. For the following analyses, an additional Cartesian coordinate system is needed. In particular, the *z*-axis is defined for each *separate* structure of the reaction in the *molecular* frame, and it points to the direction of the two newly created σ -bonds in the (usually) one-step concerted mechanism between the four involved C atoms. This is the aforementioned “reaction axis”³² of this DA reaction, i.e., the direction of main electron (density) reorganization during the reaction. An impact along the negative direction (in the definitions used here) can be conceptualized as field-augmented TS stabilization of ionic structures facilitating the formation of these bonds.^{24,25,32,86} The *y*-direction is aligned mostly along the anhydride, while the *x*-direction is orthogonal to the vertical mirror plane of all frames, showing C_s symmetry. Also illustrated is a possible EF pointing along the negative of *z*, $-F_z$, as the reaction field mostly aligned with the molecular dipole moment, which is illustrated here as a projection onto the same axis, $-\mu_z$. Note the parallel alignment of the two vectors for a *stabilization* because of the inverse sign convention, that is, the EF points from “plus” to “minus” and *vice versa* for the dipole moment (following the usual physical conventions, but differently chosen from Refs. 24, 25, and 32).

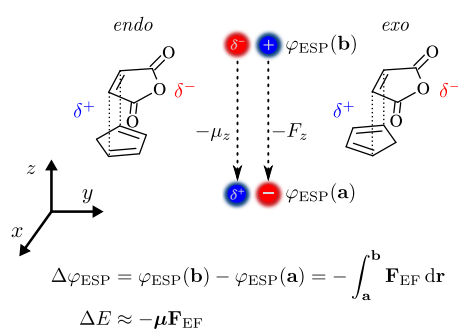


FIG. 1. Diels–Alder reaction of cyclopentadiene and maleic anhydride for both the *endo* and *exo* diastereomers including the used coordinate system definitions and the sign conventions for $\Delta\varphi_{\text{ESP}}$ as well as $\mathbf{F}_{\text{EF}}$. The molecular dipole moment lies diagonally within the molecule from the anhydride to the cyclo moiety as indicated by the $\delta^{(+,-)}$.

In the following, we will focus on the *endo* DA reaction, which is the less thermodynamically stable but faster reaction.

For later electrostatic analyses, a couple of descriptors/features were defined. Two of these are illustrated in Fig. 2. The main reaction axis ESP difference (which corresponds to a contour integral between the two given points in the conservative vector field, the EF) is measured with “ESP-TS-*z*-pl.” Another important descriptor is “ESP-TS-*y*-pl” along the *y* axis. Note that all these descriptors simply measure the electrostatic potential difference (proportional to the electric field) along the chosen directions. The GOCATs themselves can evolve to generate *any* homo- or inhomogeneous EF if it is needed for the catalysis. Hence, these directions shown here (and additional >100 of them) are not set as the input for the GOCAT optimization *a priori* but only utilized *a posteriori*, after the GA optimization, to evaluate the dominating features of the resulting EF and to investigate whether the reaction axis rule applies.

From the outset of this GOCAT optimization, the expectations with regard to literature studies of this DA reaction are clear:^{24,25,32} With an increase in the field along the reaction axis (negative of *z*), the TS is expected to be stabilized by favoring highly polarized or even charged atoms in this structure, which leads to the barrier decrease. Reaching extreme fields, a mechanistic change to a very asynchronous but still concerted mechanism is expected and, eventually, to a two-step mechanism with a zwitterionic intermediate. This ionic intermediate is the result of forming only one C–C bond first, with another subsequent TS for the second C–C bond. The C_s symmetry of the reaction is broken in this intermediate since only one bond is formed to reach the highly polarized (i.e., zwitterionic) state with formally a positive charge at the cyclopentadiene part and a negative charge at the maleic anhydride moiety. Interestingly, this development of the mechanism from the symmetric concerted one to the two-step one having an ionic intermediate at high field strengths takes place *gradually* with the increase in the field. Consequently, this leads to the asymmetric concerted mechanism in-between. Besides that, the stereoselective outcome can be controlled by subtle different barrier decreases perpendicular to this direction (cf. *y* axis), whereas a field

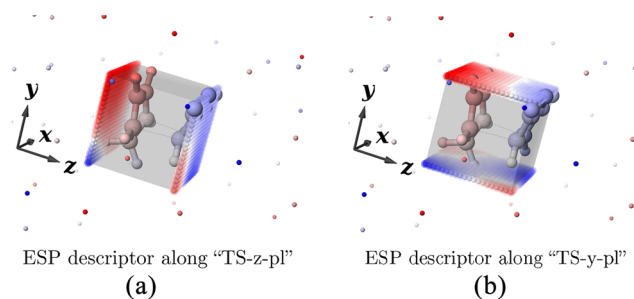


FIG. 2. Two (out of many) used descriptors for calculating an ESP difference (i.e., a highly averaged electric field) between all the “measurement” points surrounding a specific structure/frame, the TS, at a minimum bounding box. “TS-*z*-pl” (“pl” abbreviates “plane”) is hence the arithmetic average of the ESP differences of each corresponding pair along the *z*-axis, i.e., the difference along the *z*-axis, but is averaged along the *y*- and *x*-axes. The same, but *mutatis mutandis*, follows for “TS-*y*-pl.” See the depictions in the [supplementary material](#) in Sec. S1 and Fig. S1 for more information regarding the ESP descriptors.

orthogonal to the present mirror plane leads to no barrier effect at all (cf. x axis).

A. Static GOCATs on a vdW-surface or sphere

As a comparison to the effect the adaptive fitness function (*with* reaction-path relaxation) can achieve, first of all, the static GOCATs (*without* reaction-path relaxation) are investigated (cf. Algorithm S1, [supplementary material](#)). To skip the somewhat technical comparison, proceed to Sec. III B “Adaptive GOCATs on a sphere.” Due to length restrictions, additional figures and some discussions are relegated to the [supplementary material](#) of this publication (cf. Sec. S3). There, both spherical GOCATs with $N_{\text{Ch}} = 81$ and the vdW GOCATs with $N_{\text{Ch}} = 10$ that catalyze the *endo* DA reaction are illustrated. From a starting reaction barrier in the gas phase, the best GOCATs, the rank zero ones (r_0), are minimized to $\Delta E^\ddagger = 16.23 \text{ kcal mol}^{-1}$ and $\Delta E^\ddagger = 13.32 \text{ kcal mol}^{-1}$, respectively. The reference barrier of the pristine gas phase path is $\Delta E_{\text{ref}}^\ddagger = 20.08 \text{ kcal mol}^{-1}$ such that the barrier decrease is at most $\Delta\Delta E^\ddagger \geq -6.76 \text{ kcal mol}^{-1}$.

Further observations can be summarized as follows:

- The barrier effects turn out to be quite subtle in the spherical GOCATs (at most $\Delta\Delta E^\ddagger \approx -4.00 \text{ kcal mol}^{-1}$ or worse in other ranks besides the globally best found GOCAT, see Figs. S2 and S3), both in the energy domain and in its underlying ESP domain (cf. Sec. S3.1).
- GOCATs placed on a common vdW surface create more local/inhomogeneous EFs such that greater effects can be observed with an increase in flexibility and catalytic effect (see Fig. S14). At least, the best GOCAT approaches a higher symmetry, that is, a mirror plane within C_s , and also, the final ESP is somehow “structured,” showing an overall “Y” or “X” shape (Fig. S7), i.e., not just a simple field along the Cartesian z axis (cf. Sec. S3.2).
- These are all GOCAT models with enforced $\Delta E_R \leq 0 \text{ kcal mol}^{-1}$, i.e., the R frame is stabilized in the fitness function. Otherwise, GOCATs with R *destabilization* and (almost) nonexistent TS stabilization might take over in the search space. Some examples are given in Sec. S3.3 in the [supplementary material](#).
- In all cases, the main effect can be described either by the $\Delta\phi_{\text{ESP-TS-z-pl}}$ descriptor as in the spherical case or by a corresponding symmetrized difference at the attacking/attacked four C-atoms as in the vdW model (the Cartesian “plane” descriptor is not well-defined in the latter case, see the depictions of “ESP-TS-3m1-2m4” in Fig. S1 of the [supplementary material](#)). Multiple different descriptors that define more than 100 combinations of either local or rather globally averaged fields through multiple atoms of the DA molecules were screened, but the ESP differences along the Cartesian planes (“z-pl”) or at the specific reacting atoms (“3m1-2m4”) always showed the most significant correlation with the barrier decrease.
- However, especially in the vdW models, again the local ESP/EF effects might dominate in many solutions, e.g., there might be very positive ESP values at some specific H-atoms without creating the more homogeneous global trend along a certain axis (see Secs. S3.2 and S3.3 of the [supplementary material](#)).

- The greater homogeneity of the spherical GOCATs is especially present at the core atoms but not at the outer Coulomb centers: Neighboring charges tend to screen each other without notable (or with the possibility of compensating) effect at the core atoms. Hence, the Cartesian interpretability is worse in the spherical GOCATs due to this (maybe) overly extensive search space. Clearly, there are charges at different places that are almost screened by the neighbors such that no sphere with one strictly negative (red) and one strictly positive (blue) half with continuous progression in-between could be found at all. Instead, a more seemingly random blend of those charge values all around is frequent. This can be seen in Fig. S2 and below again in the adaptive version (Figs. 4 and 5).
- In general, the GOCAT solutions are not unique, i.e., several charge distributions can give rise to the same electrostatic environment, at least at the internal atoms’ positions. Note that OGOLEM, nonetheless, does not have any problems in globally optimizing meaningful ESPs/EFs. These non-unique GOCATs also have the same fitness, $f(\{\mathbf{x}_i, q_i\}_i^{N_{\text{Ch}}})$, i.e., the same catalytic effect because GOCATs with different Cartesian and charge configurations can lead to the same ESP/EF. Thus, electrostatic GOCATs might turn out both (1) to create multiple different ESP domains with substantial catalytic effects and (2) to create multiple different partial charge ensembles representing one specific ESP domain. This is generally true, including all the models used (vdW, spherical, etc), but more pronounced in the spherical GOCATs (due to their higher configurational degree of freedom). This non-uniqueness makes the analysis more difficult but also has advantages, as will be briefly discussed in Sec. IV. [As a technical side note, it can be stated that this is analogous but geometrically inverted—and more complex due to the non-linearity in the GOCAT case—to the well-known charge-fitting procedures that were already examined many years ago (see Refs. 87–89). In the latter case, optimal partial charges at (fixed) central positions in a molecule are to be found, matching a given ESP at the molecular surface. At least for molecules that are not trivially small, this is an underdetermined problem with multiple solutions. In the GOCAT case, outer charges are about to deliver an optimal catalytically active ESP at central positions, and not only the charge values but also their spatial positions are optimized, which introduces a non-linear dependence. Hence, the non-uniqueness in the GOCAT case, presumably, is even greater.]

The effects seem (at least more) subtle than the ones we expatiated upon in the first paper,²² and yet, this would translate to some (~ 2.8 – 5.0) orders of magnitude enhancement of the reaction if we bravely translate this vertical pure (electronic) energetic effect to a reaction rate at around room temperature.

B. Adaptive GOCATs on a sphere

1. Changes in energy profiles and mechanisms

In contrast to the static fitness function described in Sec. III A, the adaptive one is used in the results in the following

(cf. Algorithm 1). First, the reaction profiles of three selected GOCATs are shown in Fig. 3, which are manually selected to show a clear and already investigated^{32,34} mechanistic trend. Now, not just “vertical” energetic shifts of the profile are possible, but the structures themselves can vary strongly.

The final barriers of the GOCATs are significantly lower than those in the static case. Also the accumulated path becomes longer, which is illustrated here as summed up Euclidean distances between all the discrete frames per MEP that were found within the GOCAT, and the final frame number is dependent on the adaptive NEB creation for ensuring enough resolution of the varying path. The longer reaction coordinate stems from a greater distance between the new R and P frames caused mainly by some small rotations of these within the field and—with very strong fields—by the mechanistic change, which will be discussed below. The best GOCAT shown here for the adaptive case, r1, even has two distinct TSs. Looking at the final gradient norms, the stationary points are generally well optimized, i.e., $\|\nabla E_{[R,TS,P]}\| \ll 10 \text{ kcal mol}^{-1} \text{ \AA}^{-1}$, in contrast to the static paths, where this threshold was usually the window of “fine” gradient norms. However, with the relaxations in strong fields, and even more so when a vdW surface is used instead of the spheres because of the local ESP effects, optimizations might turn out to be either not converging at all—these would be GOCATs that would not survive during the GA—or to show slightly bigger gradients than those in the pristine gas phase at the end. Note that r1 even has almost no barrier at all, i.e., $\Delta E^\ddagger = 3.40 \text{ kcal mol}^{-1}$ (r1 was selected instead of the best GOCAT found, r0, since this r1 candidate solution shows a

slightly better resolved two-step mechanism due to the concrete NEB settings that were used; these settings were fine-tuned with regard to a compromise of performance and quality of the MEP, i.e., using small NEB-windows, not too high resolution at the end, etc. However, r0 is very similar to the selected one, r1, that is shown in the following.)

Detailed illustrations of these three GOCATs are given in Figs. 4–6 for r987, r6, and r1, respectively. The reaction path of GOCAT r987 (cf. Fig. 4) is still a textbook-style symmetric (synchronous) attack similar to the pristine gas phase reaction without any GOCAT. The distances between the respective C atoms are $d = 2.20 \text{ \AA}$ for both pairs [Fig. 4(b)]. In contrast, these distances were $d = 2.14 \text{ \AA}$ in the gas phase MEP. Interestingly, and this will be compared again, the realized fields by GOCATs become more extreme in the adaptive setting because these fields will not automatically generate large gradient norms that are just penalized, which is the case in the static fitness function without reaction-path relaxation. With an increase in the fields and a resulting stronger impact on the reaction, the main component of the field aligns better with the main reaction axis: The GOCAT generates an even better visible field along the (negative) z -direction. Such EFs are also present in the static GOCATs but with bigger fluctuations in other directions and local inhomogeneities (compare with Figs. S2 and S7 in the [supplementary material](#)). This is clearly apparent in Figs. 4(c) and 4(d): A negative ESP (red) at the diene and a positive one (blue) at the anhydride dienophile site can be seen. Moreover, the nodal surface of $\varphi_{\text{ESP}} = 0 \text{ kcal mol}^{-1} \text{ e}^{-1}$ that shows the sign change, i.e., the embedding that cuts between negative and positive ESP domains, is also pictured. (Note that the sphere and the enforced neutrality lead to the relevance of that nodal surface, which generates GOCATs with a clear zero ESP somewhere in the center. All potentials could be shifted by a constant, while the fields stay the same.) This embedding emphasizes the aforementioned observation of the z -axis aligned EF whose ESP “cuts” in the middle of the C–C-bonds, but not fully orthogonal to these newly created bonds. Moreover, this tendency is strongly amplified with respect to the static cases, where some random search space artifacts are present showing not just *one* plane-like nodal surface but a very complex embedding distant from the molecular center of the frames and hence from the atoms (Fig. S2, [supplementary material](#)).

The next GOCAT r6 already shows an unsymmetrical (asynchronous) concerted TS, but it still favors a one-step mechanism (Fig. 5). The distances between the respective C-atoms now are $d_{3m1} = 2.53 \text{ \AA}$ and $d_{2m4} = 2.14 \text{ \AA}$, with the former distance being already $\Delta d_{\text{asyn}} \approx 0.40 \text{ \AA}$ longer than in the gas phase path (cf. Fig. S1 for the C-atom labels). The field strengths are still becoming successively higher and the nodal surface becoming simpler, which is color-coded here in the same range as in r987 as well as r1 below.

Finally, r1 shows a two-step mechanism with a zwitterionic intermediate (Fig. 6). In the first TS, only one C–C-bond is formed here between the C4 and C2 atoms with $d_{2m4} = 2.20 \text{ \AA}$, while the other C atom is still very distant with $d_{3m1} = 3.10 \text{ \AA}$. After a small local minimum after the first part of the MEP (Fig. 3), the second TS then follows immediately with $d_{3m1} = 2.63 \text{ \AA}$. The field strength is one of the largest found so far while leading to converging relaxed GOCATs. These field strengths and other properties will

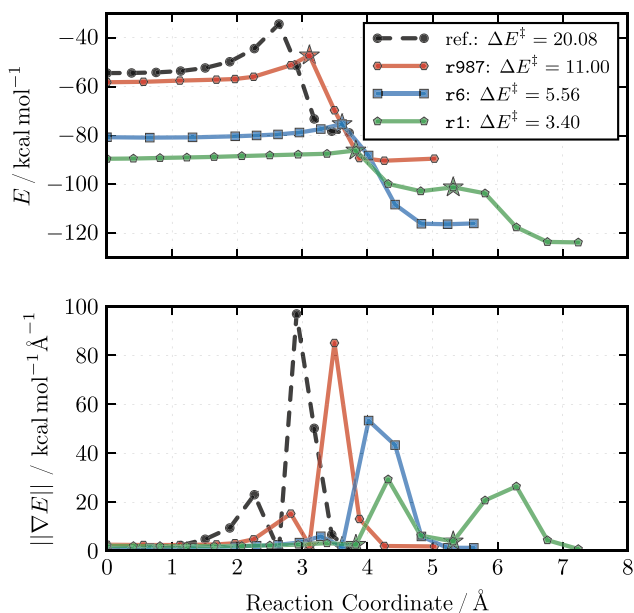


FIG. 3. $N_{\text{ch}} = 81$ (sphere, adaptive): reaction energy profiles and corresponding gradient norms of three selected *adaptive* GOCATs (ranks r1, r6, and r987) and the reference profile. The calculated barriers, $\Delta E^\ddagger$, are given in kcal mol^{-1} . The abscissa is continuous in this case since the reaction paths can vary due to the relaxations in the corresponding GOCATs. Newly found TSs are emphasized by stars.

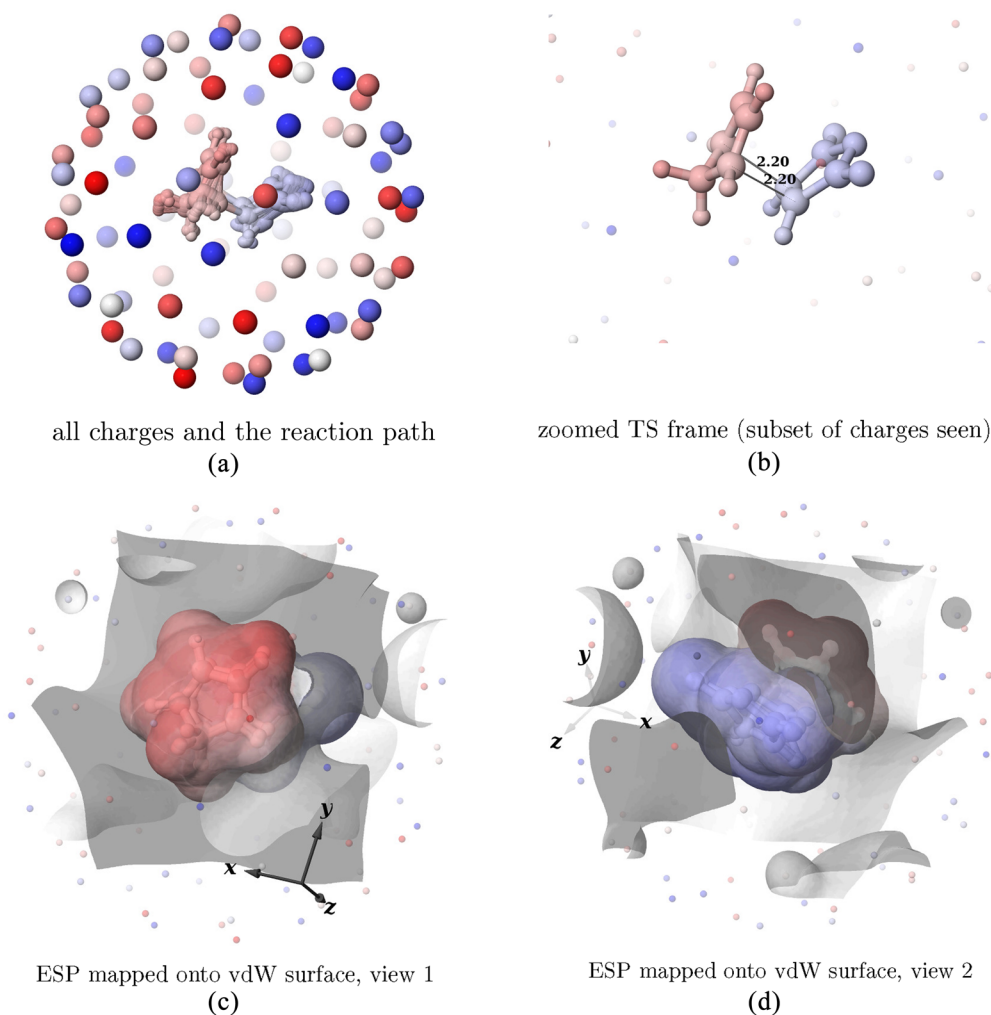


FIG. 4. $N_{\text{Ch}} = 81$ (sphere, adaptive): GOCAT r987 of Fig. 3 that stabilizes a strongly catalyzed *endo* DA reaction following a (default) synchronous concerted attack. The distances of the attacking/attacked C-atoms are given in Ångstrom in (b). φ_{ESP} values are mapped onto the common vdW surface of the three stationary frames (R, TS, and P) together with the grayish nodal surface of $\varphi_{\text{ESP}} = 0 \text{ kcal mol}^{-1} \text{ e}^{-1}$ [(c) and (d)]. The entities are colored from red to blue for $q_i \in [-1, +1] \text{ e}$ and $\varphi_{\text{ESP}} \in [-116.8, +116.8] \text{ kcal mol}^{-1} \text{ e}^{-1}$. Note that the $N_{\text{Ch}} = 81$ charges on the sphere do not follow any clearly interpretable regularity in the Cartesian domain that is given in (a); this was already discussed for the static fitness function (cf. Sec. III A).

be discussed below again, also giving some numbers then. Corresponding EF plots are also compiled in the [supplementary material](#) in Figs. S13–S17.

2. Analysis of the catalytic effects

For a more thorough inspection of a greater database after the global optimization, further scatter plots are given in Fig. 7, which also include the three selected GOCATs, r987, r6, and r1. In Fig. 7(a), the correlation of the catalytic effect expressed in the form of the resulting reaction barrier, $\Delta E^\ddagger$, and that of the averaged z -axis via the descriptor “ESP-TS- z -pl” [cf. Figure 2(a)] is clearly apparent, which is able to explain $R^2 = 0.70$ of the variance of all GOCATs. The

range of the $\Delta\varphi_{\text{ESP-TS-}z\text{-pl}}$ of up to a maximum of $\sim 100 \text{ kcal mol}^{-1} \text{ e}^{-1}$ for the most extreme EFs is also seen in the figure; the lowest barrier is created by r1 (with $\Delta E^\ddagger = 3.40 \text{ kcal mol}^{-1}$) and $\Delta\varphi_{\text{ESP-TS-}z\text{-pl}} = 86.79 \text{ kcal mol}^{-1} \text{ e}^{-1}$. Taking the distances between the measuring points into account, i.e., really measuring the averaged (due to the local inhomogeneities) field along the measure plates in Fig. 2(a), this results in $F_{\text{EF-TS-}z\text{-pl}} = -18.46 \text{ kcal mol}^{-1} \text{ Å}^{-1} \text{ e}^{-1}$ or $-1.557 \cdot 10^{-2} \text{ au}$ (in order to compare with Refs. 24 and 32).

Again, with the increase in the EF, the MEP changes are illustrated by both Figs. 7(b) and 7(c). In the former case, $\|\xi\|$ denotes the summed total reaction path length of the MEP within the GOCAT, which is the same as the maximum value of the abscissa in Fig. 3.

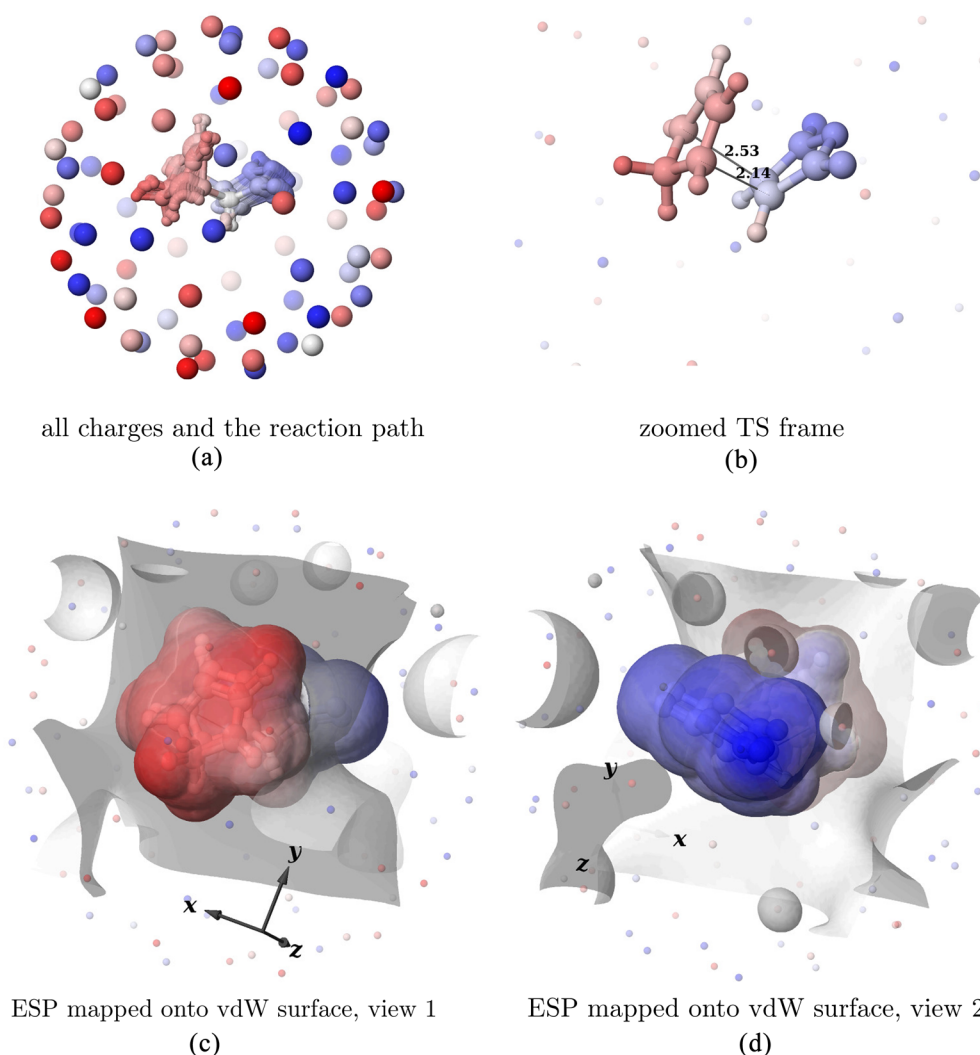


FIG. 5. $N_{\text{Ch}} = 81$ (sphere, adaptive): GOCAT r6 of Fig. 3 stabilizing a strongly catalyzed *endo* DA reaction showing an asynchronous concerted attack. The plotting details are the same as in Fig. 4.

The pristine gas phase path has $\|\xi\| = 3.76$ Å. The reason for the extended MEP length is twofold. On the one hand, the reactant and product frames re-orient within the generated oriented external EF of the GOCAT so that the orbital-controlled attack of both moieties (at the TS) includes a partially longer rotation into that configuration. Reorientation of the end-frames, i.e., of R and P, also allows for a significant stabilization of these frames. In the static fitness function, instead, the TS stabilization might already destabilize the reactant frame partially such that also the barrier effects turned out to be smaller. On the other hand, the mechanistic change of the concerted synchronous (gas phase) mechanism via an asynchronous one to the two-step mechanism takes place and is measured by Δd_{async} as the difference in distances between the one and the other CC-pair, as shown in Fig. 7(c). No difference, $\Delta d_{\text{async}} = 0$ Å,

is thus the totally symmetric case. Obviously, as indicated by the distributions at the margins of the plots, many GOCATs are still following a synchronous mechanism or only a partially asynchronous one. Yet, in the most extreme EFs, the clear spread to large Δd_{async} values is evident, i.e., with the increase in the field in the reaction axis direction, the mechanistic change is induced. This is also the second reason why the static fitness function has to show only smaller barrier effects, since the TS must change and become asymmetric, which is simply not allowed in the static fitness function and thus results in large gradients and, consequently, large fitness penalties. Interestingly, these features and tendencies are included in the search space of possible candidate solutions *without* any further bias from the outset. Note that these findings are in line with the ones of Ref. 32 (also reported in Refs. 24 and

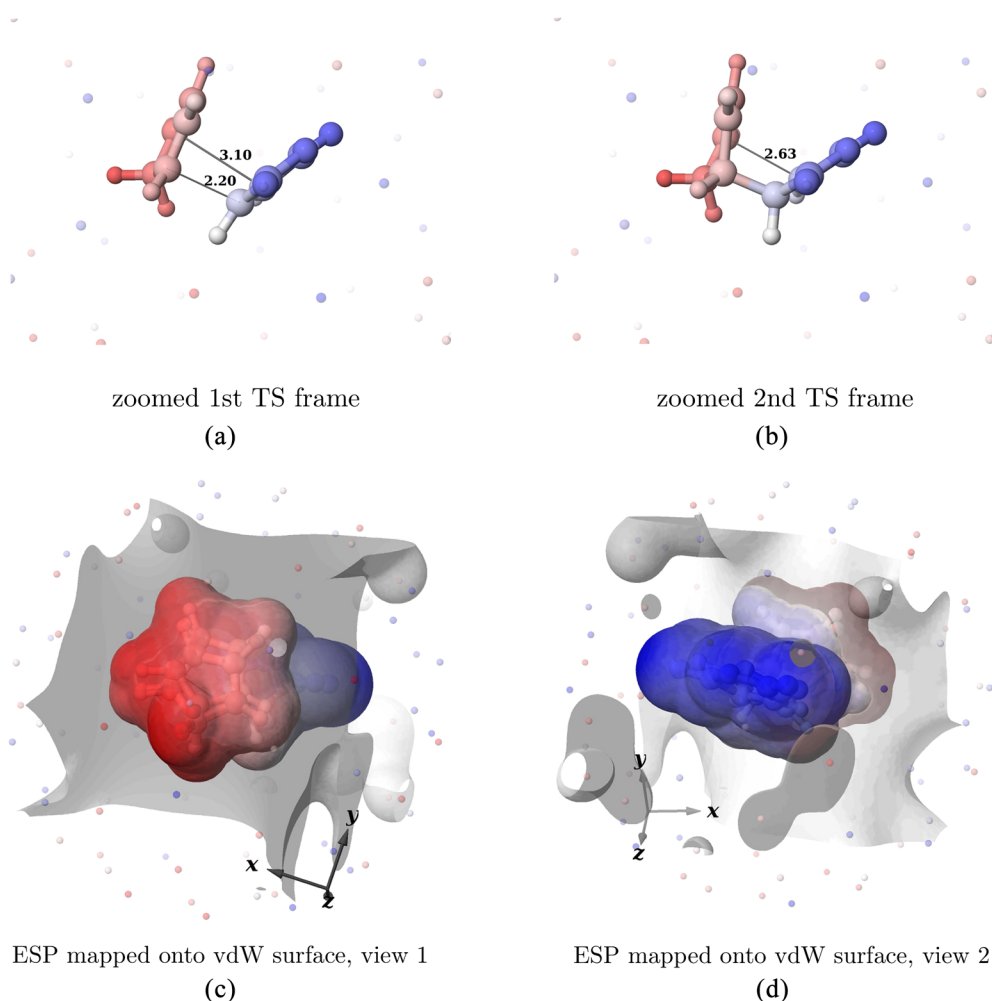


FIG. 6. $N_{\text{Ch}} = 81$ (sphere, adaptive): GOCAT r1 of Fig. 3 stabilizing a strongly catalyzed *endo* DA reaction showing an asynchronous two-step mechanism via a zwitterionic intermediate. The plotting details are the same as in Fig. 4. Note that a corresponding figure of all the GOCAT charges as shown in Figs. 4(a) and 5(a) is not given here due to the worse interpretability in the Cartesian domain as discussed in the main text.

25 and extended upon in Ref. 34). Besides, mechanism variants of DA reactions, from synchronous concerted to two-step ones, are usual organic chemistry textbook knowledge, and these mechanisms depend on the specific functional groups of the reacting moieties.⁹⁰ However, these mechanistic variations are elicited here by the EFs rather than by the electronic influence of different substitution patterns.

In Ref. 32, for instance, a perfectly uniform EF of about $F_z = -1.25 \cdot 10^{-2} \text{ au} = -14.8 \text{ kcal mol}^{-1} \text{ \AA}^{-1} \text{ e}^{-1}$ was applied and, with the help of an additional implicit solvent in the strongest fields, the two-step mechanism with a clear minimum was reported (which is then better stabilized by the solvent). The difference to the GOCAT setting can be traced back to the differences between the levels of theory, i.e., density functional theory (DFT) in Ref. 32 vs semi-empirical PM7, the (strict) uniformity of the field in

Ref. 32, and the *one* GOCAT that must serve to influence the *complete* MEP at once vs (at least) the frame-wise changing implicit solvent in Ref. 32.

However, the GOCATs used in the present contribution are free to evolve as needed in order to induce the catalytic effect. That is, no direction whatsoever needs to be defined and, after the general complexity of the resulting EF is controlled to a preset range by the meta-parameters of the GOCAT model, the *maximum* electrostatic catalytic effect can be worked out by *global* optimization. Hence, also other components than in the reaction axis direction can develop if they serve the purpose of catalysis. This is illustrated in Fig. 8. Now, the ESP difference (i.e., the scaled EF), in the *y*-direction is plotted [cf. Figure 2(b)]. This figure contrasts the GOCAT effects on the *endo* with the ones on the *exo* distereomer, which was not yet done anywhere in this article (and

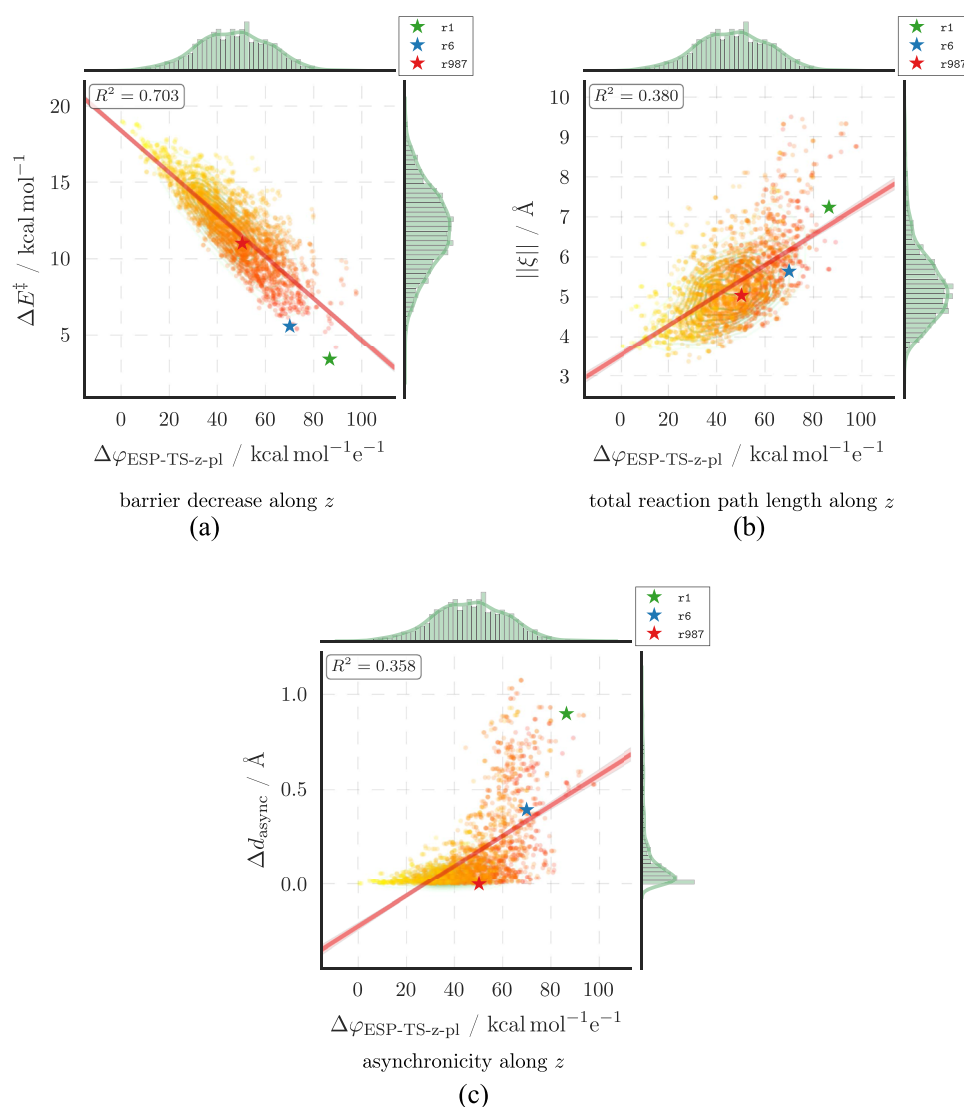


FIG. 7. $N_{\text{Ch}} = 81$ (sphere, adaptive): correlations of $N_{\text{GOCAT}} = 2952$ adaptive GOCATs with mean $\|\nabla E\| < 5 \text{ kcal mol}^{-1} \text{\AA}^{-1}$ along the z-axis for the *endo* case. $\Delta E^\ddagger$ is the final (highest) energy barrier of each separate GOCAT shown, $\|\xi\|$ is the summed reaction path length, and Δd_{async} is the absolute difference between the two CC-pairs at the TS frame when the two new bonds are formed. For each case, scatter (correlation) plots are shown using a color bar between yellow/red for worse/good fitness values as well as histograms at the margins. The three, already discussed GOCATs, r1, r6 and r987, are marked explicitly.

also not in the ESI). The only difference between the diastereomers is the impact a y -axis field can have. Obviously, in the *endo* case [cf. Fig. 8(a)], there is a spread of EFs in both positive and negative directions around $\Delta\varphi_{\text{ESP-TS-y-pl}} = 0 \text{ kcal mol}^{-1} \text{e}^{-1}$ but no clear (or only a miniscule) correlation at all, despite the seeming but accidental trend of only the three selected ranks, r987, r6, and r1. In contrast, in Fig. 8(b), the *exo* GOCAT database indeed has evolved many GOCATs that follow more clearly a trend with regard to stronger EFs along the y -axis. In Ref. 32, the EF effects along the orthogonal Cartesian directions were studied separately, and a notable bigger impact onto the *exo* diastereomer was reported. In the GOCAT framework, this feature emerges on its own, while the GOCAT is allowed to develop components in any direction, including (if needed) local inhomogeneities, i.e., even emulating parts of the additional solvent mentioned in

Ref. 32. In Refs. 24, 32, and 34, it was demonstrated that this possible discrimination of the diastereomers could then induce diastereoselectivity in the present reaction. Note that the results in Fig. 8 are still based on two completely separate global optimization runs. That is, we have not (yet) set up a pool of GOCAT individuals that are allowed to influence both the *endo* and *exo* MEP within one and the same global optimization. In this way, a fitness function would also be imaginable which directly encodes a maximum stereoselectivity and evolves promising features to exactly amplify that.

Complementary to the discussions above, a more quantitative comparison between the static GOCATs, the adaptive GOCATs, completely uniform EFs [“plate capacitor (GO)CATs”], and the literature is given in the [supplementary material](#) (especially in Secs. S5 and S6).

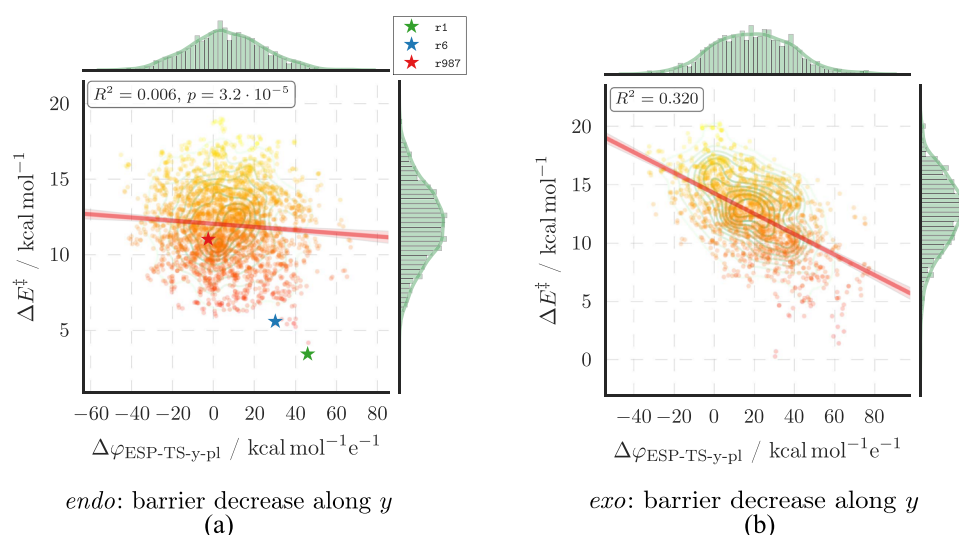


FIG. 8. $N_{\text{Ch}} = 81$ (sphere, adaptive): correlations of $N_{\text{GOCAT}} = 2952$ adaptive GOCATs with mean $\|\nabla E\| < 5$ kcal mol $^{-1}$ Å $^{-1}$ along the y -axis for the *endo* case in Fig. 8(a) (*endo* was shown in all figures so far). For comparison, correlations for $N_{\text{GOCAT}} = 1776$ adaptive GOCATs with mean $\|\nabla E\| < 5$ kcal mol $^{-1}$ Å $^{-1}$ along the y -axis for the *exo* diastereomer are given in (b).

IV. CONCLUSIONS AND OUTLOOK

In our previous paper,²² we had introduced our “globally optimal catalysts” (GOCATs) framework that allows us to create optimized catalysts for any given, elementary reaction step. By stepping back from the real but vast chemical space into an abstract but comparatively very small representation space of GOCAT ingredients, we can afford to employ non-deterministic global optimization in such way that high probabilities for finding good global optimum candidates are achievable in practice.

In the present work, we have still restricted ourselves to simple point charges as only GOCAT ingredients, but we have lifted a critical restriction: In our previous work, we had to stick to a pre-defined reaction path during GOCAT optimization. Since the GOCAT is meant to influence the reaction it surrounds, this is unrealistic and was a main cause for the comparatively small catalytic effects obtained before.²² Now, we have included a robust on-the-fly reaction path optimization into the procedure, allowing for more drastic impacts of the GOCAT onto the potential energy surface, in particular, including substantial GOCAT-induced modifications of the reaction path, all the way to full-blown mechanistic changes.

To demonstrate that this extended GOCAT method is far superior to its predecessor and that despite the chemically abstract point charges, the resulting catalytic trends can be well understood and thus are highly useful, this improved framework was applied to a standard Diels–Alder reaction. Previous experimental and theoretical work had already established that this reaction is catalyzed by z -axis-oriented electric fields by multiple orders of magnitude, up to a mechanistic change from the default synchronous concerted mechanism to a step-wise (non-pericyclic) one, and that electric fields in y -axis-orientation influence the stereoselectivity (cf. Fig. 1 for the definition of the coordinate axes relative to the reactants). All these catalytic tendencies re-emerged in our present GOCAT study without using any *a priori* information in any of these respects.

Of course, the electric fields generated by our spatially distributed point charges (and influencing the reacting molecules) are

meant to model the electric fields generated by the effective partial charges (electron density and nuclear charges) of the molecular framework of an actual catalyst. Hence, the present work solely addressed the optimization of these fields via the optimization of the distributed point charges [“step (A)” in the introduction]; finding a real molecular catalyst that approximates and realizes such an optimized electric field [“step (B)”] still is an open problem. However, as indicated above, analyzing the catalytic effects of an optimized, abstract GOCAT already produces important insights into catalytic mechanisms that are helpful guides in the search of good, real catalysts. The non-uniqueness of the GOCAT configuration with regard to specific ESP domains can now turn out to be a clear advantage of having multiple translation targets: Not only it is possible for a real molecular catalyst to only approximately realize the optimal GOCAT-prescribed electric field while still achieving a substantial catalytic effect but also it is possible to find such real-molecular approximations for *several* of the best (but different) GOCATs with almost equal fitness, and then, a real-molecular catalyst approximating a GOCAT solution that is formally not the globally optimal one might even turn out to be better in the real-molecule world than the one approximating the globally optimal GOCAT.

Nevertheless, “step (B)” still remains a challenge. However, we do believe that most of the tools needed for this step are already established. Computational design of molecular systems to optimize molecular properties by now is an accepted part of theoretical chemistry.¹ Normally, properties to be optimized by design are functions of atomic positions only in a very indirect way (via these atomic positions entering into the electronic Schrödinger equation, from which the electronic wavefunction and the property in question are calculated). Hence, it appears to be a reasonable idea to adopt such design methods to the much more directly accessible target of several partial charges at given positions (i.e., a GOCAT result), optimally approximated by suitable atoms with their suitable partial charges, linked by an essentially charge-neutral backbone or cage. During such an optimizing translation from an abstract GOCAT into a real

molecular catalyst framework, navigating chemical compound space can be greatly facilitated by alchemical gradients,⁹¹ turning the discrete problem into a continuous one, while still retaining the ability to project back into the discrete space of real molecules with real atoms, even during such optimizations.⁹² Work along these lines is under progress in our lab.

Note that our GOCATs can only provide a catalytic electric field that is optimal within the pre-set restrictions (finite number of point charges, allowed charge values, and allowed Cartesian space); the truly optimal field may require more point charges with different values. A more subtle effect still missing in our “step (A)” GOCAT method is that our point charges are not polarizable, in contrast to a real molecular catalyst. Hence, the external catalytic electric fields that we optimized in our GOCATs are the effective end results after a catalyst has polarized the reacting molecules and vice versa, which makes “step (B)” more challenging. Likewise, we are still aiming at *one* optimized GOCAT point charge arrangement for the whole course of the reaction, while the atomic positions in a true molecular catalyst likely also respond dynamically to the changing atomic positions in the reacting molecules. To only allow one “frozen” GOCAT while the reaction is now able to change is one of the further fundamental assumptions of the present model that we have already critically discussed in Ref. 22. This assumption will also be lifted in the future to converge to a more realistic setting.

Of course, the GOCAT concept can be easily extended beyond partial point charges by implementing also other abstract (or more concrete atomic/molecular) entities for all types of non-bonded (and bonded) interactions.

As indicated at the end of Sec. III B 2, an interesting further improvement would be to incorporate multiple different reaction paths including stereoselective discrimination, etc., and the precise control of such reaction alternatives. In the present case, especially when using spheres instead of unique vdW surfaces, after minor additional implementations, both diastereomers (or enantiomers, with regard to Ref. 34) could then be optimized synchronously with arbitrary targets in the fitness function. These additional implementations would also need to address intrinsic measures of different reaction mechanisms that could occur during the optimizations, e.g., by graph-based description of reaction networks and similarity measures to discern more subtle TS changes.

Last but not least, true catalysis requires the modeling of complete multi-step catalytic cycles, possibly with the exchange of electrons, protons, hydrogen atoms, etc., and definitely with the regeneration of the catalyst at the end. All the above improvement avenues were already discussed and listed in our previous paper²² and will be the subject of future work in our group.

SUPPLEMENTARY MATERIAL

See the [supplementary material](#) for more information on the electrostatic catalysis evaluations, more GOCATs stemming from a static/vertical fitness function for different GOCAT models, detailed illustrations and tables for the electrostatic scalar and vector fields that are generated, and an in-depth comparison of all of these, including the literature.

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